Basic and Advanced Portfolio Optimization

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Learning Objectives and Topic

Goal: to present a self-contained treatment of portfolio optimization, with an eye toward applications

1. Why Optimize? Why Mean-Variance?
2. How do we use factor models in MVO?
3. How do we add a factor to the model?
4. How does the Information Ratio relate to IC and diversification?
5. When is centralized equivalent to decentralized portfolio management?
6. How do constraints and estimation errors affect portfolio construction?

Assumptions:

1. You have been exposed to some mean-variance optimization
2. Some mathematical maturity

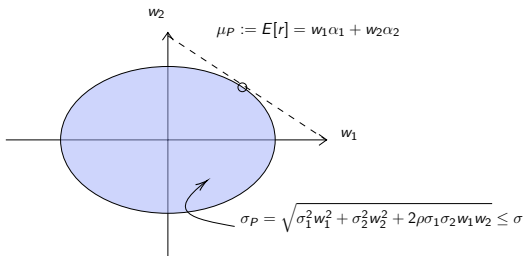
Reference Book: **Elements of Quantitative Investing**. Paleologo2025)

The MVO Optimal Portfolio is Sharpe Maximizing

Assume $\mathbf{\Omega_r} \succ 0$, $\boldsymbol{\alpha} \neq 0$ and unrestricted positions. The following problems have the same optimal direction:

- Maximize portfolio PnL for a maximum portfolio variance.
- $\Leftrightarrow$ Maximize portfolio PnL for a target portfolio risk (std. dev.)
- $\Leftrightarrow$ Maximize portfolio PnL per unit of portfolio risk

$$\max_{\mathbf{w}} \boldsymbol{\alpha}^T \mathbf{w} \quad \text{s.t. } \sqrt{\mathbf{w}^T \mathbf{\Omega_r} \mathbf{w}} \leq \sigma, \quad \sigma > 0$$



- The risk constraint binds when maximizing expected PnL.
- Sharpe is scale-invariant: a Sharpe-maximizing portfolio can be scaled to this risk budget.

Closed form solution of the optimal portfolio

Maximize the Lagrangian (multiplier $\lambda \geq 0$)

$$\mathcal{L}(\mathbf{w}, \lambda) = \boldsymbol{\alpha}^T \mathbf{w} - \lambda \left(\mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r} \mathbf{w} - \sigma^2 \right).$$

Setting the gradient in $\mathbf{w}$ to zero: $\boldsymbol{\alpha} - 2\lambda \boldsymbol{\Omega}_\mathbf{r} \mathbf{w} = 0 \implies \mathbf{w}^\star = \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha} / (2\lambda)$.
Minimizing the dual $g(\lambda) = \frac{\boldsymbol{\alpha}^T \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha}}{4\lambda} + \lambda \sigma^2$ over $\lambda > 0$ gives

Solution: Return and Risk are Balanced under Optimality

At the optimum, marginal expected return equals λ times marginal variance: $\boldsymbol{\alpha} = 2\lambda \boldsymbol{\Omega}_\mathbf{r} \mathbf{w}$.

$$\mathbf{w}^\star = \frac{\sigma}{\sqrt{\boldsymbol{\alpha}^T \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha}}} \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha} \qquad \lambda^\star = \frac{\sqrt{\boldsymbol{\alpha}^T \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha}}}{2\sigma}$$

Volatility is σ; Sharpe is $\sqrt{\boldsymbol{\alpha}^T \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha}}$. Why is λ absent from $\mathbf{w}^\star$?

Worked Example: Computing the Optimal Portfolio

Three assets with diagonal covariance. Returns and volatilities are annual; weights are fractions of NAV and residual cash earns the risk-free rate.

Inputs

- $\boldsymbol{\alpha} = (0.12,\ 0.08,\ 0.04)^\top$
- $\sigma = (0.25,\ 0.20,\ 0.15)$
- $\boldsymbol{\Omega}_{\mathbf{r}} = \mathrm{diag}(\sigma_i^2)$
- Vol budget $\sigma_P = 0.10$

Precision-weighted alphas ($\boldsymbol{\Omega}_{\mathbf{r}}^{-1}\boldsymbol{\alpha}$)

- $[\boldsymbol{\Omega}_{\mathbf{r}}^{-1}\boldsymbol{\alpha}]_1 = 0.12/0.0625 = 1.92$
- $[\boldsymbol{\Omega}_{\mathbf{r}}^{-1}\boldsymbol{\alpha}]_2 = 0.08/0.0400 = 2.00$
- $[\boldsymbol{\Omega}_{\mathbf{r}}^{-1}\boldsymbol{\alpha}]_3 = 0.04/0.0225 = 1.78$

Maximum Sharpe Ratio

$$\mathrm{SR}^\star = \sqrt{\boldsymbol{\alpha}^\top \boldsymbol{\Omega}_{\mathbf{r}}^{-1} \boldsymbol{\alpha}}$$

$$= \sqrt{0.2304 + 0.1600 + 0.0711} \approx 0.679$$

Normalized weights

$$w_i^\star = \frac{\sigma_P}{\mathrm{SR}^\star} [\boldsymbol{\Omega}_{\mathbf{r}}^{-1}\boldsymbol{\alpha}]_i$$

- $w_1^\star \approx 0.283$
- $w_2^\star \approx 0.294$
- $w_3^\star \approx 0.262$

Observation: Asset 1 has the highest alpha and Sharpe (0.48) but less weight than asset 2: here $w_i \propto s_i/\sigma_i$, and asset 1 is the most volatile.

MVO Unconstrained Formulation

Equivalent unconstrained (penalized) problem, with $\lambda > 0$:

$$\max_{\mathbf{w}} \boldsymbol{\alpha}^T \mathbf{w} - \lambda \mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r} \mathbf{w}$$

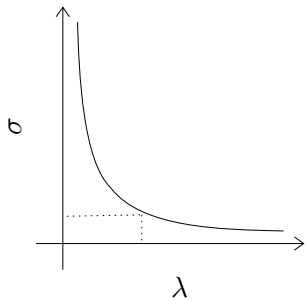
Solution:

$$\mathbf{w}^\star = \frac{1}{2\lambda} \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha}$$

Same as vol-constrained solution when

$$\lambda = \frac{\sqrt{\boldsymbol{\alpha}^T \boldsymbol{\Omega}_\mathbf{r}^{-1} \boldsymbol{\alpha}}}{2\sigma}$$

Figure: λ translates portfolio variance into a return measure.



Pricing out the constraint

The penalty coefficient λ is the shadow price per unit of variance at the corresponding constrained optimum.

How does the Sharpe of the portfolio change with λ?

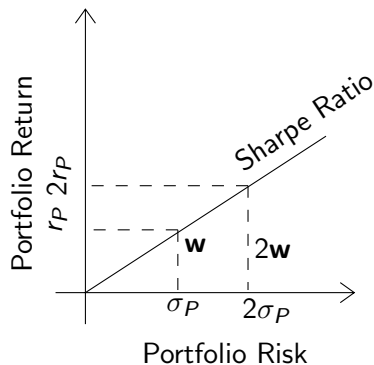
Maximum Sharpe Ratio

- For $\mathbf{w} = \mathbf{\Omega_r}^{-1}\boldsymbol{\alpha}$, PnL is $\mathbf{w}^T\boldsymbol{\alpha} = \boldsymbol{\alpha}^T\mathbf{\Omega_r}^{-1}\boldsymbol{\alpha}$
- Portfolio risk: $\sqrt{\mathbf{w}^T\mathbf{\Omega_r}\mathbf{w}} = \sqrt{\boldsymbol{\alpha}^T\mathbf{\Omega_r}^{-1}\boldsymbol{\alpha}}$

Maximum Sharpe Ratio:

$$\text{SR}^\star = \sqrt{\boldsymbol{\alpha}^T\mathbf{\Omega_r}^{-1}\boldsymbol{\alpha}}$$

Figure: Portfolio Risk and Return Scale Linearly



Factor Risk Model I

A factor is a broad, persistent driver of return.

- **Broad**: Exists in different regions, countries, sectors, and industries (in multiple asset classes).
- **Persistent**: Persists over time, across business cycles.

Factor exposures can be constructed using characteristics of the stock; factor returns may be observed or estimated. ($\mathbf{b}_i \in \mathbb{R}^n \qquad \mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \cdots \mathbf{b}_m]$).

- A **factor model** decomposes asset returns into a *systematic* component driven by m common factors and an *idiosyncratic* residual:

$$\mathbf{r} = \mathbf{B}\mathbf{f} + \boldsymbol{\epsilon}, \qquad \text{Cov}(\boldsymbol{\epsilon}, \mathbf{f}) = 0$$

- $\mathbf{B} \in \mathbb{R}^{n \times m}$ is the **factor loading matrix**: column j gives every asset's exposure to factor j.
- $\mathbf{f} \in \mathbb{R}^m$ are the **factor returns**, which may be observed or estimated.

Factor Risk Model II

- $\boldsymbol{\epsilon} \in \mathbb{R}^n$ is **idiosyncratic risk** — the part of each asset's return unexplained by the factors.

Covariance decomposition. Taking the covariance of both sides, and using $\text{Cov}(\boldsymbol{\epsilon}, \mathbf{f}) = 0$:

$$\boldsymbol{\Omega}_\mathbf{r} = \underbrace{\mathbf{B}\,\boldsymbol{\Omega}_\mathbf{f}\,\mathbf{B}^T}_{\text{systematic}} + \underbrace{\boldsymbol{\Omega}_\boldsymbol{\epsilon}}_{\text{idiosyncratic}}$$

where $\boldsymbol{\Omega}_\mathbf{f} = \text{cov}(\mathbf{f})$ is the $m \times m$ factor covariance and $\boldsymbol{\Omega}_\boldsymbol{\epsilon} = \text{cov}(\boldsymbol{\epsilon})$ is assumed **diagonal** ($\boldsymbol{\Omega}_\boldsymbol{\epsilon} = \begin{bmatrix} \sigma^2_{\epsilon_1} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma^2_{\epsilon_n} \end{bmatrix}$).

Factor-Mimicking Portfolios (FMPs) I

- A **factor** is a broad driver of return. Factor exposures can be estimated using stock characteristics ($\mathbf{b} \in \mathbb{R}^n$).
- An **FMP** is the portfolio that has *unit exposure* to one factor, zero exposure to the others, and minimum idiosyncratic variance among portfolios with these exposures.

Assume $\mathbf{\Omega}_{\boldsymbol{\epsilon}} \succ 0$ and $\mathbf{B}$ has full column rank. Given $\mathbf{r} = \mathbf{B}\mathbf{f} + \boldsymbol{\epsilon}$,
$\mathbf{B} = \left[\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_m\right] \in \mathbb{R}^{n \times m}$.
FMP for factor i: minimum-idio-variance portfolio with unit exposure to factor i:

$$\min_{\mathbf{w}} \ \mathbf{w}^T \mathbf{\Omega}_{\boldsymbol{\epsilon}} \mathbf{w} \qquad \text{s.t.} \quad \mathbf{B}^T \mathbf{w} = \mathbf{e}_i$$

Factor-Mimicking Portfolios (FMPs) II

Solution. Using the Lagrangian with multiplier $\boldsymbol{\nu} \in \mathbb{R}^m$, the first-order condition gives $2\boldsymbol{\Omega}_\epsilon \mathbf{w} = \mathbf{B}\boldsymbol{\nu}$, so $\mathbf{w} = \frac{1}{2}\boldsymbol{\Omega}_\epsilon^{-1}\mathbf{B}\boldsymbol{\nu}$. Substituting into the constraint $\mathbf{B}^T\mathbf{w} = \mathbf{e}_i$ fixes $\boldsymbol{\nu} = 2(\mathbf{B}^T\boldsymbol{\Omega}_\epsilon^{-1}\mathbf{B})^{-1}\mathbf{e}_i$. The matrix of *all* FMPs (columns = individual FMPs) is:

$$\boxed{\mathbf{P} = \boldsymbol{\Omega}_\epsilon^{-1}\mathbf{B}\,(\mathbf{B}^T\boldsymbol{\Omega}_\epsilon^{-1}\mathbf{B})^{-1}} \in \mathbb{R}^{n\times m}, \qquad \mathbf{P}^T\mathbf{B} = \mathbf{I}_m$$

Why Optimize?

- "Why optimize" is a very deep question. For example, do we optimize in everyday decisions? In general, no (see, e.g. *Simple Heuristics that Make us Smart*, Gigerenzer2016)).
- Do *engineers* optimize? No, when they invent something. Yes, when they refine something. Often, they *satisfy constraints*
- Do *economists* optimize? Yes, unfortunately that's all they do
- Should we optimize?
  - We have a well-defined, narrow domain, data and some (broad) *preferences*
  - We are prescriptive, but want to perform well in real life
  - So **we optimize**, but **with caveats**.
  - Recurrent theme: understand the limits of your formulation. We can always be wrong

When Optimization Is and Is Not Appropriate

- Optimization is best suited when the domain is *well-specified*:
  - Objectives can be quantified (risk, return, cost)
  - Constraints are stable and interpretable
  - Data volume justifies the model complexity
- It is *fragile* when used to substitute for judgement:
  - Garbage-in / garbage-out: optimizing a noisy signal amplifies noise
  - Over-fitted solutions fail out-of-sample (Michaud1989))
  - Extreme portfolios are a symptom of a poorly specified problem, not a solution

Rule of thumb:

The optimizer is a *calculator*, not a decision-maker. Maximize what you genuinely believe; minimize what you do not.

Linear Algebra

Vectors

Weight vector:

$\mathbf{w} = [w_1, \ldots, w_n]^\top \in \mathbb{R}^n$,

Expected Returns

$\boldsymbol{\mu} := \mathbb{E}[\mathbf{r}]$

Covariance Matrix

$\boldsymbol{\Omega} := \mathbb{E}[(\mathbf{r} - \boldsymbol{\mu})(\mathbf{r} - \boldsymbol{\mu})^\top] \in \mathbb{R}^{n \times n}$

$\boldsymbol{\Omega}$ is symmetric positive semi-definite;

portfolio variance $\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Omega} \mathbf{w}$

portfolio expected returns $\mu_p = \mathbf{w}^\top \boldsymbol{\mu}$

Inner Products

$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$

Cauchy–Schwarz: $|\mathbf{u}^\top \mathbf{v}| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$

Key Linear Algebra Tools

- **Cholesky:** $\boldsymbol{\Omega} = \mathbf{L}\mathbf{L}^\top$ for $\boldsymbol{\Omega} \succ 0$
- **SVD**: $\boldsymbol{\Omega} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^\top$
- **Eigen-decomposition** $\boldsymbol{\Omega} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top, \quad \boldsymbol{\Omega}\mathbf{u} = \lambda \mathbf{u}$
- Linear model $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon}$, with $E[\boldsymbol{\epsilon}] = 0$ and $\mathrm{Cov}(\mathbf{x}, \boldsymbol{\epsilon}) = 0$
- Mean: $\mathbb{E}[\mathbf{y}] = \mathbf{A}\mathbb{E}[\mathbf{x}]$
- Variance $\boldsymbol{\Omega}_y = \mathbf{A}\boldsymbol{\Omega}_x\mathbf{A}^T + \boldsymbol{\Omega}_\epsilon$
- **Woodbury identity:** $(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}$; converts $n \times n$ inverses to $m \times m$ when $m \ll n$

Matrix Calculus & Bayesian Statistics

Matrix Calculus Identities

Expression	$\partial/\partial\mathbf{w}$
$\mathbf{a}^\top\mathbf{w} = \mathbf{w}^\top\mathbf{a}$	$\mathbf{a}$
$\mathbf{w}^\top\mathbf{A}\mathbf{w}$	$(\mathbf{A}+\mathbf{A}^\top)\mathbf{w}$
$\mathbf{w}^\top\mathbf{A}\mathbf{w}$ ($\mathbf{A}$ symmetric)	$2\mathbf{A}\mathbf{w}$
$\frac{1}{2}\mathbf{w}^\top\boldsymbol{\Omega}\mathbf{w} - \boldsymbol{\mu}^\top\mathbf{w}$ (MVO obj.)	$\boldsymbol{\Omega}\mathbf{w} - \boldsymbol{\mu}$
$\mathrm{tr}(\mathbf{A}\mathbf{U})$ (w.r.t. $\mathbf{U}$)	$\mathbf{A}^\top$
$\log\lvert\mathbf{A}\rvert$ (w.r.t. $\mathbf{A}$)	$(\mathbf{A}^{-1})^\top$

Convention: $\partial f/\partial\mathbf{w} = \nabla f$ is a column vector (gradient, or denominator, layout).

Chain rule, with J_g the Jacobian of g:
$\nabla(f \circ g) = J_g^\top \nabla f$
More at
https://www.matrixcalculus.org/

Bayesian Statistics Bayes' Theorem

$$\underbrace{p(\theta \mid x)}_{\text{posterior}} \propto \underbrace{p(x \mid \theta)}_{\text{likelihood}} \cdot \underbrace{p(\theta)}_{\text{prior}}$$

Key Concepts

- **Prior** $p(\theta)$: beliefs *before* observing data
- **Likelihood** $p(x \mid \theta)$: data generating model
- **Posterior** $p(\theta \mid x)$: updated beliefs, summarized by mean & covariance
- **Conjugate prior:** posterior in the same family as prior $\Rightarrow$ closed-form update

Portfolio Management

Mean Variance

Why Mean-Variance?

Mean–variance analysis can be motivated by preferences or by the information used in a decision.

- Utility: consider an increasing, concave one-period wealth utility V.
  - For gains small relative to 1/(absolute risk aversion), a second-order expansion gives mean–variance.
  - Exponential utility with normally distributed gains gives an exact mean–variance objective.

  See, e.g., Huang and Litzenberger1988).
- Information: if asset returns are characterized by their means and variances, or investors make decisions based on mean and variances alone
  - Mean–variance comparisons can apply to elliptical returns when the relevant moments exist and preferences satisfy suitable conditions.
  - Higher moments can be estimated, but estimates are often noisy and require stronger moment assumptions.
- Additional considerations:
  - Sharpe Ratio maximization is a special case of MVO.
  - MVO fits factor models like a glove

Utility Maximization

Let W_0 denote the terminal value of initial wealth invested at the risk-free rate, and let $\mathbf{w}$ be dollar positions in risky assets. Then terminal wealth is:

$$\boxed{W = W_0 + \mathbf{w}^T \mathbf{R} - \sum_i w_i R_f}$$

- Investment is represented as a vector $\mathbf{w}$
- Total return is represented by the vector $\mathbf{R}$
- Borrow at risk-free rate R_f
- Excess returns are $\mathbf{r} = \mathbf{R} - R_f \mathbf{1}$

Goal: choose $\mathbf{w}$ to maximize

$$\max_{\mathbf{w}} E[V(W)] = \max_{\mathbf{w}} E\left[V(W_0 + \mathbf{w}^T \mathbf{r})\right], \quad \text{where } V \text{ is increasing and concave.}$$

Mean-Variance as Local Utility Optimization

$$\boldsymbol{\mu}_{\mathbf{r}} := E[\mathbf{r}] \quad \boldsymbol{\Omega}_{\mathbf{r}} := E[(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{r}}) \cdot (\mathbf{r} - \boldsymbol{\mu}_{\mathbf{r}})^T]$$

For sufficiently smooth V, expand $V(W_0 + \mathbf{w}^T\mathbf{r})$ around W_0, treating $\mathbf{w}^T\mathbf{r}$ as a small perturbation and assuming the expected remainder is negligible: Baron1977); Benveniste et al.2024)

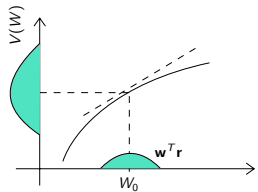
$$V(W_0 + \mathbf{w}^T\mathbf{r}) = V(W_0) + V'(W_0)\,(\mathbf{w}^T\mathbf{r}) + \tfrac{1}{2}V''(W_0)\,(\mathbf{w}^T\mathbf{r})^2 + \underbrace{O\big[(\mathbf{w}^T\mathbf{r})^3\big]}_{\text{drop}}$$

Take expectations, using $E[\mathbf{w}^T\mathbf{r}] = \mathbf{w}^T\boldsymbol{\mu}_{\mathbf{r}}$ and $E[(\mathbf{w}^T\mathbf{r})^2] = \mathbf{w}^T\boldsymbol{\Omega}_{\mathbf{r}}\mathbf{w} + (\mathbf{w}^T\boldsymbol{\mu}_{\mathbf{r}})^2$.
Dropping the squared mean requires the additional approximation $(\mathbf{w}^T\boldsymbol{\mu}_{\mathbf{r}})^2 \ll \mathbf{w}^T\boldsymbol{\Omega}_{\mathbf{r}}\mathbf{w}$:

$$E[V(W)] \approx \underbrace{V(W_0)}_{\text{constant}} + V'(W_0)\,\mathbf{w}^T\boldsymbol{\mu}_{\mathbf{r}} + \tfrac{1}{2}V''(W_0)\,\mathbf{w}^T\boldsymbol{\Omega}_{\mathbf{r}}\mathbf{w}$$

If $V'(W_0) > 0$ and $V''(W_0) < 0$, define absolute risk aversion:

$$\rho := -\frac{V''(W_0)}{V'(W_0)} > 0.$$



From Utility to Mean-Variance

The objective becomes: **Mean-Variance Optimization** (MVO).

$$\max_{\mathbf{w}} \quad \underbrace{\mathbf{w}^T \boldsymbol{\mu}_\mathbf{r}}_{\text{expected return}} - \frac{\rho}{2} \underbrace{\mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r} \mathbf{w}}_{\text{portfolio variance}}$$

MVO is the consequence of maximizing expected utility under the assumptions:

- Portfolio gains are small on the curvature scale of V at W_0, and
- Terms beyond second order and the squared expected gain are negligible.
- Note: The risk-aversion parameter ρ is a property of the investor, not of the market.

Exponential and Log Utility

Exponential utility: $V(x) = -e^{-\rho x}$

- Absolute risk aversion $= \rho$ (constant, independent of wealth)
- If gains are normally distributed, maximizing expected utility is equivalent to maximizing $\mathbf{w}^T \boldsymbol{\mu}_\mathbf{r} - \frac{\rho}{2}\mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r}\mathbf{w}$.

Log utility (Kelly criterion): $V(x) = \log(x)$

- Absolute risk aversion $= 1/W_0$ (decreasing in wealth)
- A local approximation gives $\max_{\mathbf{w}} \mathbf{w}^T \boldsymbol{\mu}_\mathbf{r} - \frac{1}{2W_0}\mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r}\mathbf{w}$ when the squared mean is negligible.
- Higher wealth lowers *absolute* risk aversion; relative risk aversion is constant.
- Log utility requires strictly positive terminal wealth.

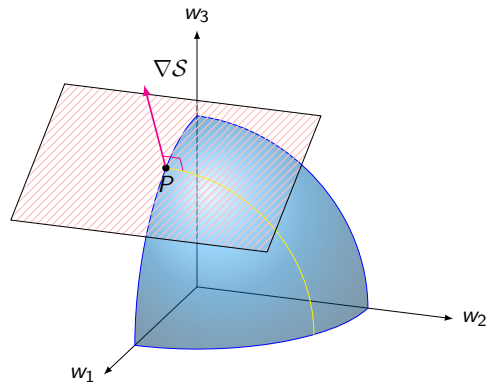
The MVO objective: Geometric Interpretation

- Given expected returns $\boldsymbol{\alpha} \in \mathbb{R}^n$
- and covariance $\boldsymbol{\Omega}_\mathbf{r} \in \mathbb{R}^{n \times n}$,
- maximize expected PnL
- subject to a volatility budget $\sigma > 0$:

$$\max_{\mathbf{w}} \boldsymbol{\alpha}^T \mathbf{w}$$
$$\text{s.t. } \mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r} \mathbf{w} \leq \sigma^2$$

- $\mathcal{C} : \boldsymbol{\alpha}^T \mathbf{w} - C = 0$ is a hyperplane at distance $|C|/\|\boldsymbol{\alpha}\|$ from the origin
- $\mathcal{S} : \mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r} \mathbf{w} \leq \sigma^2$ is an ellipsoid
- $\mathcal{C}$ is tangent to $\mathcal{S}$ when $\boldsymbol{\alpha} = \lambda \nabla_\mathbf{w} (\mathbf{w}^T \boldsymbol{\Omega}_\mathbf{r} \mathbf{w})$

Figure: Visualizing the objective



Risk Aversion and Volatility Targeting

For dollar positions $\mathbf{w}$, consider the penalized problem $\max_{\mathbf{w}} \boldsymbol{\alpha}^T\mathbf{w} - \lambda\mathbf{w}^T\boldsymbol{\Omega}_\mathbf{r}\mathbf{w}$. In the local utility approximation, $\lambda = \rho/2$.

- With $\boldsymbol{\Omega}_\mathbf{r} \succ 0$ and $\boldsymbol{\alpha} \neq 0$, set λ to attain a desired volatility $\sigma^\star$:

$$\lambda = \frac{\sqrt{\boldsymbol{\alpha}^T\boldsymbol{\Omega}_\mathbf{r}^{-1}\boldsymbol{\alpha}}}{2\sigma^\star}.$$

- The required penalty depends on the forecast alpha scale and the covariance model.
- A drawdown target requires a model of returns and an investment horizon. Expected maximum drawdown is not a worst-case bound or a drawdown quantile.

Key Properties of MVO Solution

- $\mathbf{w}^\star \propto \mathbf{\Omega_r}^{-1}\boldsymbol{\alpha}$ (precision matrix times alpha)
- For expected-return maximization with a binding volatility budget and no other constraints or penalties, portfolio size is proportional to σ.
- In that same problem, multiplying all alphas by $\kappa > 0$ leaves the optimal portfolio unchanged.

Alpha scale and portfolio weights

A common positive rescaling changes expected return and the shadow price, but not the weights in this volatility-budget problem. This invariance need not hold with other constraints or penalties.

$\lambda^\star$ is the **shadow price** of the variance budget: a small relaxation $d(\sigma^2)$ increases optimal expected return by approximately $\lambda^\star d(\sigma^2)$.

Thinking in terms of Volatility

Hypothetical example: \$100M invested in each of NVDA and GE has different risk.

MVO from Sharpe Ratios and Correlations.

Let $\mathbf{V} = \text{diag}(\sigma_1, \ldots, \sigma_n)$ (asset volatilities), $\mathbf{C}$ (correlation matrix). Then $\mathbf{\Omega_r} = \mathbf{VCV}$.

Define:

- $\mathbf{v}^\star := \mathbf{V}\mathbf{w}^\star$ (optimal \$ vols)
- $\mathbf{s}$ (vector of Sharpe, $s_i = \alpha_i/\sigma_i$)

	NVDA	GE
Annual vol σ_i	50%	20%
Alpha α_i (ann.)	10%	4%
Sharpe $s_i = \alpha_i/\sigma_i$	0.20	0.20
\$ Volatility	\$50M	\$20M

Solution in volatility space:

Optimal dollar volatilities depend on Sharpe Ratios and the inverse correlation matrix.

$$\mathbf{v}^\star = \frac{1}{2\lambda}\mathbf{C}^{-1}\mathbf{s} \qquad \text{and} \qquad \text{SR}^\star = \sqrt{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}$$

Heuristic: the optimal dollar volatilities are, to a first approximation, proportional to asset Sharpe ratios

Example: MVO from Sharpe Ratios and Correlations (3 Assets)

Setup. All figures use a common annual horizon; all assets have $\sigma_i = 0.20$. Pairwise correlation: $\rho_{12} = 0.80$, $\rho_{13} = \rho_{23} = 0.10$.

Asset Sharpe Ratios s

	α_i	$s_i = \alpha_i/\sigma_i$
Asset 1	0.12	0.60
Asset 2	0.08	0.40
Asset 3	0.06	0.30

Correlation matrix C

$$\mathbf{C} = \begin{bmatrix} 1.00 & 0.80 & 0.10 \\ 0.80 & 1.00 & 0.10 \\ 0.10 & 0.10 & 1.00 \end{bmatrix}$$

Precision matrix $\mathbf{C}^{-1}$

$$\mathbf{C}^{-1} \approx \begin{bmatrix} 2.78 & -2.22 & -0.06 \\ -2.22 & 2.78 & -0.06 \\ -0.06 & -0.06 & 1.01 \end{bmatrix}$$

Optimal dollar-vol allocations $\mathbf{v}^\star \propto \mathbf{C}^{-1}\mathbf{s}$:

$$v_1^\star \propto 2.78(0.60) - 2.22(0.40) - 0.06(0.30) = +\mathbf{0.76}$$
$$v_2^\star \propto -2.22(0.60) + 2.78(0.40) - 0.06(0.30) = -\mathbf{0.24}$$
$$v_3^\star \propto -0.06(0.60) - 0.06(0.40) + 1.01(0.30) = +\mathbf{0.25}$$

Special case: If all assets are uncorrelated, the maximum portfolio Sharpe ratio is the square root of the sum of squared asset Sharpe ratios.

Example: MVO from Sharpe Ratios and Correlations (3 Assets)

Exact vs. Heuristic (dollar-vol allocations normalized to $\sum_i |v_i| = 1$):

	Asset 1	Asset 2	Asset 3	SR*
Heuristic $\mathbf{v} \propto \mathbf{s}$	$+0.46$	$+0.31$	$+0.23$	0.59
MVO exact $\mathbf{v} \propto \mathbf{C}^{-1}\mathbf{s}$	$+0.61$	$\mathbf{-0.19}$	$+0.20$	**0.66**

- Asset 2 has a *positive* Sharpe (0.40) yet MVO **shorts it**: its alpha is low relative to asset 1, with which it is highly correlated ($\rho_{12} = 0.80$). The short position hedges shared risk.
- Asset 3 has the *lowest* Sharpe (0.30), yet its long position has a similar absolute size to the short position in asset 2. Its correlations with the other assets are low ($\rho \approx 0.10$).
- The heuristic loses about 10% of optimal Sharpe in this example. The loss depends on both correlations and the direction of $\mathbf{s}$.

Quiz-I: Error in Heuristic

Error in Heuristic

What would be the error in Sharpe when using the heuristic?

Hint: Think about the alignment of $\mathbf{s}$ with the eigenvectors of $\mathbf{C}$.

- $\mathbf{C} = \mathbf{I}$ gives zero Sharpe loss.
- Correlation need not cause a loss if $\mathbf{s}$ is an eigenvector of $\mathbf{C}$.

Quiz-II: Correlations, Dispersion, and Limits to Performance

When all assets have pairwise correlation ρ:

- What are the possible values of ρ?
- What is the optimal vol allocation?
- What is the optimal Sharpe ratio?
- If all the assets have the same Sharpe ratios equal to s, what is the Sharpe ratio of the optimal portfolio?
- What is the asymptotic behavior in the number of assets?

Reading the Precision Matrix

Let $\mathbf{\Theta} = \mathbf{\Omega}_{\mathbf{r}}^{-1}$ and write $\rho_{ij|-ij}$ for the partial correlation between the residuals from linearly projecting assets i and j on the remaining assets. Under joint normality, this is also the conditional correlation. Then

$$w_i^\star \propto \Theta_{ii}\alpha_i + \sum_{j \neq i} \Theta_{ij}\alpha_j = \Theta_{ii}\left(\alpha_i - \sum_{j \neq i} \rho_{ij|-ij}\sqrt{\frac{\Theta_{jj}}{\Theta_{ii}}}\,\alpha_j\right).$$

- $\Theta_{ij} = -\rho_{ij|-ij}\sqrt{\Theta_{ii}\Theta_{jj}}$.
- The contribution of another asset depends on its alpha and its partial correlation with asset i.
- Strong correlations can produce large offsetting positions; the result also depends on the expected returns.

Intuition: Why the Precision Matrix?

- The inverse covariance matrix accounts for how each position changes portfolio risk in combination with the others.
- With uncorrelated assets, optimal weights are proportional to alpha divided by variance.
- Highly correlated assets can generate offsetting positions when their alphas differ. At perfect correlation the covariance matrix is singular and the inverse formula no longer applies.
- Small covariance eigenvalues amplify differences in estimated alphas and can make the solution sensitive to estimation error.

Factor Risk Models: Efficient Computation

Why this structure matters

With n stocks and $m \ll n$ factors, $\mathbf{\Omega_r}$ is *low-rank plus diagonal*. Assume $\mathbf{\Omega_\epsilon} \succ 0$ and $\mathbf{\Omega_f} \succ 0$. The Woodbury identity then gives $\mathbf{\Omega_r^{-1}} = \mathbf{\Omega_\epsilon^{-1}} - \mathbf{\Omega_\epsilon^{-1}B}(\mathbf{\Omega_f^{-1}} + \mathbf{B}^T\mathbf{\Omega_\epsilon^{-1}B})^{-1}\mathbf{B}^T\mathbf{\Omega_\epsilon^{-1}}$, reducing the leading cost of applying the inverse from dense $n \times n$ inversion to operations of order $O(nm^2 + m^3)$ when $\mathbf{\Omega_\epsilon}$ is diagonal.

Whitening of Returns: Dividing Out Idiosyncratic Risk I

- When idiosyncratic variances differ, high-variance assets can dominate the noise in an unweighted cross-sectional least-squares or quadratic analysis.
- **Whitening** rescales returns so that every asset contributes equally in idiosyncratic-risk units:

$$\tilde{\mathbf{r}} := \boldsymbol{\Omega}_{\epsilon}^{-1/2}\,\mathbf{r} \quad \Longleftrightarrow \quad \tilde{r}_i = \frac{r_i}{\sigma_{\epsilon_i}}$$

Effect on the factor model. Applying $\boldsymbol{\Omega}_{\epsilon}^{-1/2}$ to $\mathbf{r} = \mathbf{B}\mathbf{f} + \boldsymbol{\epsilon}$:

$$\boldsymbol{\Omega}_{\epsilon}^{-1/2}\mathbf{r} = \underbrace{\boldsymbol{\Omega}_{\epsilon}^{-1/2}\mathbf{B}}_{\tilde{\mathbf{B}}}\,\mathbf{f} + \underbrace{\boldsymbol{\Omega}_{\epsilon}^{-1/2}\boldsymbol{\epsilon}}_{\tilde{\boldsymbol{\epsilon}}}$$

The whitened residual $\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\Omega}_{\epsilon}^{-1/2}\boldsymbol{\epsilon}$ has covariance $\mathbf{I}$. Its components have unit variance and are uncorrelated; independence requires additional distributional assumptions.

Factor-Mimicking Portfolios (Connection to Whitening)

Interpretation

- Column i of $\mathbf{P}$ is the FMP for factor i: precision-weighted projection of factor i onto return space.
- $\mathbf{P}^T\mathbf{B} = \mathbf{I}_m$: each FMP has *exactly* unit exposure to its own factor and zero to all others.

Connection to whitening

- $\mathbf{P} = \mathbf{\Omega}_\epsilon^{-1}\mathbf{B}\,\mathbf{H}^{-1}$ where $\mathbf{H} = \mathbf{B}^T\mathbf{\Omega}_\epsilon^{-1}\mathbf{B}$ is the whitened information matrix.
- The GLS factor fit is $\hat{\mathbf{f}} = \mathbf{P}^T\mathbf{r}$. Equivalently regress $\mathbf{\Omega}_\epsilon^{-1/2}\mathbf{r}$ on $\mathbf{\Omega}_\epsilon^{-1/2}\mathbf{B}$. It is conditionally unbiased for $\mathbf{f}$ if $E[\epsilon \mid \mathbf{f}] = 0$.

Separating Factor and Residual Alpha

For $\gamma > 0$, consider $\max_{\mathbf{w}} \boldsymbol{\alpha}^T\mathbf{w} - (2\gamma)^{-1}\mathbf{w}^T\boldsymbol{\Omega}_\mathbf{r}\mathbf{w}$. Let $\mathbf{H} = \mathbf{B}^T\boldsymbol{\Omega}_\epsilon^{-1}\mathbf{B}$ and $\mathbf{P} = \boldsymbol{\Omega}_\epsilon^{-1}\mathbf{B}\mathbf{H}^{-1}$. Decompose alpha and portfolio as

$$\boldsymbol{\alpha} = \mathbf{B}\boldsymbol{\mu} + \boldsymbol{\alpha}_\perp, \quad \mathbf{B}^T\boldsymbol{\Omega}_\epsilon^{-1}\boldsymbol{\alpha}_\perp = 0, \qquad \mathbf{w} = \mathbf{P}\mathbf{u} + \mathbf{v}, \quad \mathbf{B}^T\mathbf{v} = 0.$$

Then $\mathbf{P}^T\boldsymbol{\alpha}_\perp = 0$ and $\mathbf{P}^T\boldsymbol{\Omega}_\epsilon\mathbf{v} = 0$. The mean–variance objective separates exactly:

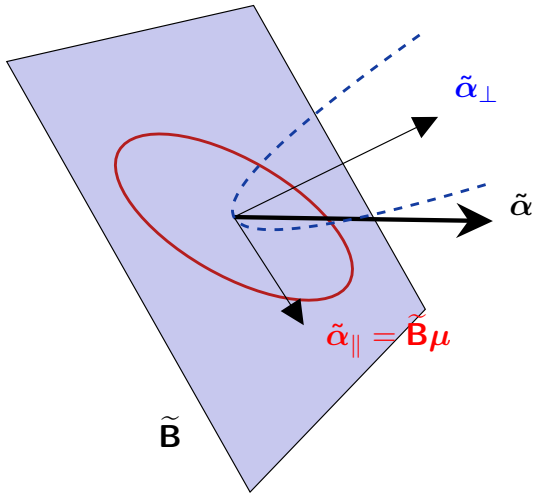
$$\underbrace{\boldsymbol{\mu}^T\mathbf{u} - \frac{1}{2\gamma}\mathbf{u}^T(\boldsymbol{\Omega}_\mathbf{f} + \mathbf{H}^{-1})\mathbf{u}}_{\text{factor portfolio}} + \underbrace{\boldsymbol{\alpha}_\perp^T\mathbf{v} - \frac{1}{2\gamma}\mathbf{v}^T\boldsymbol{\Omega}_\epsilon\mathbf{v}}_{\text{factor-neutral alpha}}.$$

The factor portfolio retains idiosyncratic risk $\mathbf{H}^{-1}$.

Geometry of the Factor and Alpha Decomposition

In whitened coordinates, let $\widetilde{\mathbf{B}} = \boldsymbol{\Omega}_{\epsilon}^{-1/2}\mathbf{B}$ and $\tilde{\boldsymbol{\alpha}} = \boldsymbol{\Omega}_{\epsilon}^{-1/2}\boldsymbol{\alpha}$.

- The factor component $\widetilde{\mathbf{B}}\boldsymbol{\mu}$ lies in the span of the whitened loadings.
- The residual component $\boldsymbol{\Omega}_{\epsilon}^{-1/2}\boldsymbol{\alpha}_{\perp}$ is orthogonal to that span.
- The idiosyncratic risk metric becomes Euclidean after whitening.



Optimization in Factor Space

When expected returns are explained by factor exposures, $\boldsymbol{\alpha} = \mathbf{B}\boldsymbol{\mu}$, the factor portfolio has expected return $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Omega}_\mathbf{f} + \mathbf{H}^{-1}$:

$$\mathbf{u}^\star = \gamma(\boldsymbol{\Omega}_\mathbf{f} + \mathbf{H}^{-1})^{-1}\boldsymbol{\mu}, \qquad \mathbf{w}^\star = \mathbf{P}\mathbf{u}^\star.$$

The maximum Sharpe ratio is $\sqrt{\boldsymbol{\mu}^T(\boldsymbol{\Omega}_\mathbf{f} + \mathbf{H}^{-1})^{-1}\boldsymbol{\mu}}$. If the FMPs are sufficiently diversified, $\mathbf{H}^{-1}$ may be small relative to $\boldsymbol{\Omega}_\mathbf{f}$; neglecting it gives the simpler factor-only approximation.

Adding a New Factor

Given loading matrix $\mathbf{B}$, residual covariance $\mathbf{\Omega}_\epsilon \succ 0$, and a new characteristic vector $\mathbf{a}$, remove its existing loading exposure in the idiosyncratic-risk metric:

$$\mathbf{b}_{m+1} = \left[\mathbf{I} - \mathbf{B}(\mathbf{B}^T\mathbf{\Omega}_\epsilon^{-1}\mathbf{B})^{-1}\mathbf{B}^T\mathbf{\Omega}_\epsilon^{-1}\right]\mathbf{a}.$$

If $\mathbf{b}_{m+1} \neq 0$, its factor-mimicking portfolio is

$$\mathbf{v}_{m+1} = \frac{\mathbf{\Omega}_\epsilon^{-1}\mathbf{b}_{m+1}}{\mathbf{b}_{m+1}^T\mathbf{\Omega}_\epsilon^{-1}\mathbf{b}_{m+1}}.$$

It has unit exposure to the new loading and zero exposure to the existing loadings. Its idiosyncratic variance is $(\mathbf{b}_{m+1}^T\mathbf{\Omega}_\epsilon^{-1}\mathbf{b}_{m+1})^{-1}$.

- This orthogonalization does not make factor returns uncorrelated.
- Estimating the new factor premium and covariances can change existing allocations.

Trading in Idiosyncratic Space

Alpha Orthogonal Portfolio

The residual $\boldsymbol{\alpha}_\perp$ is the $\boldsymbol{\Omega}_\epsilon^{-1}$-weighted cross-sectional projection residual of expected returns on the factor loadings.

Pure-alpha portfolio (no factor exposures):

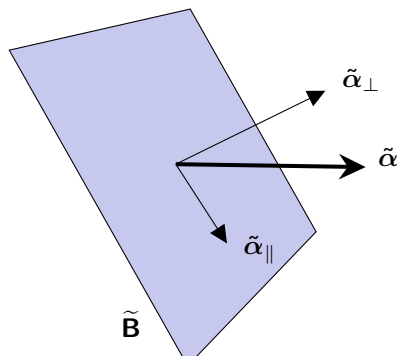
$$\max_{\mathbf{w}} \boldsymbol{\alpha}^T \mathbf{w}$$
$$\text{s.t.} \mathbf{B}^T \mathbf{w} = 0, \quad \mathbf{w}^T \boldsymbol{\Omega}_\epsilon \mathbf{w} \leq \sigma^2$$

Solution: Orthogonalize, then inverse-variance weight:

$$\boldsymbol{\alpha}_\perp := \left(\mathbf{I}_n - \mathbf{B}(\mathbf{B}^T \boldsymbol{\Omega}_\epsilon^{-1} \mathbf{B})^{-1} \mathbf{B}^T \boldsymbol{\Omega}_\epsilon^{-1}\right) \boldsymbol{\alpha}$$
$$\mathbf{w}^\star = \frac{\sigma}{\sqrt{\boldsymbol{\alpha}_\perp^T \boldsymbol{\Omega}_\epsilon^{-1} \boldsymbol{\alpha}_\perp}} \boldsymbol{\Omega}_\epsilon^{-1} \boldsymbol{\alpha}_\perp$$

Assume $\boldsymbol{\alpha}_\perp \neq 0$; otherwise no factor-neutral alpha is available.

Figure: Orthogonalization in whitened coordinates



Information Coefficient

For this example, let $\mathbf{r}$ denote residual returns and suppose $\mathbf{r} = \boldsymbol{\alpha} + \boldsymbol{\epsilon}$, with $E[\boldsymbol{\epsilon}] = 0$ and $\mathrm{Cov}(\boldsymbol{\epsilon}) = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}} \succ 0$. Let $\boldsymbol{\eta} \neq 0$ be a fixed forecast used to form $\mathbf{w} = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}}^{-1}\boldsymbol{\eta}$, and define

$$\tilde{\mathbf{r}} = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}}^{-1/2}\mathbf{r}, \quad \tilde{\boldsymbol{\alpha}} = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}}^{-1/2}\boldsymbol{\alpha}, \quad \tilde{\boldsymbol{\eta}} = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}}^{-1/2}\boldsymbol{\eta}.$$

$$E[\mathrm{PnL}] = \tilde{\boldsymbol{\alpha}}^{T}\tilde{\boldsymbol{\eta}}, \qquad \mathrm{Vol}(\mathrm{PnL}) = \|\tilde{\boldsymbol{\eta}}\|, \qquad \mathrm{SR} = \frac{\tilde{\boldsymbol{\alpha}}^{T}\tilde{\boldsymbol{\eta}}}{\|\tilde{\boldsymbol{\eta}}\|}.$$

For I uniform on $\{1, \ldots, n\}$ and independent of $\boldsymbol{\epsilon}$, define the *pooled* $\mathrm{IC} = \mathrm{Corr}(\tilde{\eta}_I, \tilde{r}_I)$. Assume the whitened signal and alpha are cross-sectionally centered.

The Fundamental Law of Active Management

Under the preceding model, $\mathrm{Cov}(\tilde{\epsilon}) = \mathbf{I}_n$. With centered $\tilde{\boldsymbol{\eta}}$ and $\tilde{\boldsymbol{\alpha}}$ and uniform index I,

$$\mathrm{IC} = \frac{\tilde{\boldsymbol{\alpha}}^T \tilde{\boldsymbol{\eta}}}{\|\tilde{\boldsymbol{\eta}}\| \sqrt{n + \|\tilde{\boldsymbol{\alpha}}\|^2}}, \qquad \mathrm{SR} = \mathrm{IC}\sqrt{n + \|\tilde{\boldsymbol{\alpha}}\|^2}.$$

If average squared signal is small relative to residual variance, $\|\tilde{\boldsymbol{\alpha}}\|^2 / n \ll 1$, the familiar approximation is

$$\boxed{\mathrm{SR} \approx \mathrm{IC}\sqrt{n}}, \qquad \mathrm{SR}_{\mathrm{annual}} \approx \mathrm{IC}\sqrt{nT}$$

when the T period P&Ls have the same mean and variance and are uncorrelated. Breadth is nT in this idealized example. If P&L is active return relative to a benchmark and its volatility is tracking error, SR is an information ratio. Dependence, constraints and costs require additional adjustments. In a centered scalar regression with an intercept, $R^2 = \mathrm{IC}^2$.

Fundamental Law: Numerical Examples

Scenario A: Daily cross-sectional model

- $n = 500$ stocks, daily rebalancing ($T = 252$)
- IC $= 0.02$ (a "weak" signal)
- SR $= 0.02 \times \sqrt{500 \times 252} \approx 7.1$
- This assumes 500 independent names and 252 uncorrelated periods.

Scenario B: Macro discretionary PM

- $n = 10$ macro themes, monthly ($T = 12$)
- IC $= 0.20$ (very skilled)
- SR $= 0.20 \times \sqrt{10 \times 12} \approx 2.2$
- Skill compensates for low breadth, but only to a point.

Key tensions

- Trading more often can raise market impact and reduce net returns.
- Expanding the universe can change average forecast quality.
- Costs, correlations, and forecast errors can reduce realized SR relative to the idealized calculation.

Common misuse

The law is often cited to justify adding more strategies or assets. Independent bets provide the full nominal breadth. Correlated bets may still help, but their contribution depends on their signals and covariance.

Transfer Coefficient: Portfolio Geometry

For fixed true alpha $\boldsymbol{\alpha} \neq 0$, covariance $\boldsymbol{\Omega}_{\boldsymbol{\epsilon}} \succ 0$, and nonzero portfolio $\mathbf{w}$, let

$$\mathbf{u} = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}}^{-1/2}\boldsymbol{\alpha}, \qquad \mathbf{v} = \boldsymbol{\Omega}_{\boldsymbol{\epsilon}}^{1/2}\mathbf{w}, \qquad \mathrm{TC}_{\mathrm{geom}} := \frac{\mathbf{u}^T\mathbf{v}}{\|\mathbf{u}\|\ \|\mathbf{v}\|}.$$

Then the portfolio Sharpe ratio and the unconstrained optimum satisfy

$$\mathsf{SR}(\mathbf{w}) = \mathrm{TC}_{\mathrm{geom}}\ \mathsf{SR}^{\star}, \qquad \mathsf{SR}^{\star} = \|\mathbf{u}\|, \quad |\mathrm{TC}_{\mathrm{geom}}| \leq 1.$$

This is a cosine in risk-whitened coordinates. It is a Pearson correlation only when the vectors are centered in the relevant cross-sectional sense. True alpha is not observed; a coefficient based on forecasts may differ.

Transfer Coefficient and Forecast Error

If $\mathbf{\Omega}_{\epsilon} = \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)$, TC_{geom} is the cosine between (α_i/σ_i) and $(w_i\sigma_i)$.

- A forecast-based optimizer may use $\hat{\boldsymbol{\alpha}}$ in place of $\boldsymbol{\alpha}$; its population Sharpe depends on alignment with the *true* alpha, not just on the forecast.
- Against the true unconstrained optimum, the geometric coefficient cannot exceed one.
- The cross-sectional transfer coefficient in Clarke et al.2002) measures a related, but not generally identical, alignment.

Transfer Coefficient: Case Studies

Clarke et al.2002, Table 1, p. 53) study an S&P 500 benchmark, a 5% active-risk target and $\mathrm{IC} = 0.067$. Their benchmark fundamental-law estimate is $0.067\sqrt{500} \approx 1.50$.

Constraint set	**TC**	**Estimated IR**
Baseline (full investment)	0.98	1.47
Long-only	0.58	0.87
Long-only + market-cap neutral	0.47	0.70
Multiple constraints	0.31	0.46

Multiple constraints include long-only, market-cap and dividend-yield neutrality, and a 50% turnover limit. IRs are ex ante fundamental-law estimates.

Quiz-III: The Correlation Chain Rule

Let r be the realized residual return, α the forecast, and $\hat{\alpha}$ the alpha implied by the implemented portfolio, with $\mathrm{IC} = \mathrm{Corr}(r, \alpha)$ and $\mathrm{TC} = \mathrm{Corr}(\alpha, \hat{\alpha})$:

$$\underbrace{r}_{\text{realized}} \longmapsto \underbrace{\alpha}_{\text{forecast}} \longmapsto \underbrace{\hat{\alpha}}_{\text{implemented}}$$

- Under what conditions is $\mathrm{Corr}(r, \hat{\alpha}) = \mathrm{IC} \times \mathrm{TC}$?
- Give an example where the identity fails.

Estimation Error and Constraints

- With known parameters, a constraint cannot improve the optimum over a larger feasible set. With estimated parameters, it can reduce sensitivity to estimation error and improve out-of-sample Sharpe.
- Long-only and gross-exposure limits can limit extreme allocations, but also exclude potentially valuable positions.
- Shrinking a covariance estimate toward a scaled identity adds an ℓ_2 term to a quadratic risk penalty, up to rescaling; this is one way to regularize estimated risk.
- Whether any constraint or shrinkage method helps is an empirical question that depends on the portfolio and the estimation problem.

Quiz-IV: Relationship between Hit Rate and IC

Let a scalar forecast α and return r be centered and jointly normal, with positive variances and Pearson correlation IC. The hit rate is $h = P(\alpha r > 0)$. Show that

Hit Rate and IC

$$h = \frac{1}{2} + \frac{\arcsin(\mathrm{IC})}{\pi}.$$

The result also holds at $\mathrm{IC} = \pm 1$ by continuity.

Decentralized vs. Centralized

Let $\mathbf{A} = [\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_k]$ contain the PMs' signals and let $\boldsymbol{\Omega}_\mathbf{r} \succ 0$ be a shared risk matrix. Set $\mathbf{W} = \boldsymbol{\Omega}_\mathbf{r}^{-1}\mathbf{A}$.

- **Centralized:** combine signals as $\boldsymbol{\alpha} = \mathbf{A}\mathbf{u}$ and optimize: $\mathbf{w}_{\text{cen}} \propto \boldsymbol{\Omega}_\mathbf{r}^{-1}\mathbf{A}\mathbf{u} = \mathbf{W}\mathbf{u}$.
- **Decentralized:** form the columns of $\mathbf{W}$ separately, then combine them with weights $\mathbf{u}$: $\mathbf{w}_{\text{dec}} = \mathbf{W}\mathbf{u}$.
- Each PM's position can be rescaled if its combining weight is adjusted by the inverse rescaling.

Combining sleeve portfolios by their own Sharpes requires expected sleeve returns and covariances under the same aggregate alpha model.

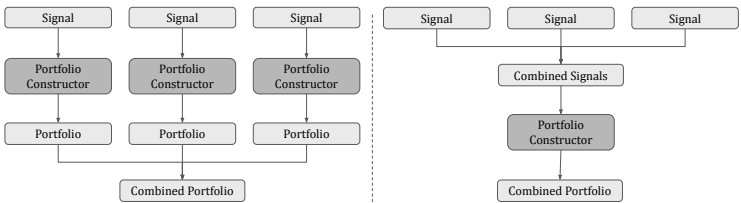
Centralized vs. Decentralized: Frictions

The equivalence uses a shared covariance matrix, unconstrained linear MVO, consistent signal-combination weights, and no transaction costs. It can fail when:

- PMs use different risk models or different position and leverage limits;
- trading costs depend on the net combined position or vary by desk;
- sleeves use nonlinear rules or the combining weights differ from the weights used to aggregate signals.

A central optimizer can account for netting and shared exposures. Combining portfolio outputs can still be useful, but depends on the information and constraints available to the allocator.

Equivalence Result



With the same $\mathbf{\Omega_r}$, no frictions, and matching signal weights, $\mathbf{\Omega_r}^{-1} \sum_i u_i \boldsymbol{\alpha}_i = \sum_i u_i \mathbf{\Omega_r}^{-1} \boldsymbol{\alpha}_i$. Thus linear MVO commutes with signal aggregation.

Foundations and Active Management: Key Takeaways

1. MVO is justified by interpretability, tractability, and utility-theoretic foundations
2. Optimal portfolio: $\mathbf{w}^\star \propto \mathbf{\Omega}_\mathbf{r}^{-1}\boldsymbol{\alpha}$; Sharpe: $\mathrm{SR}^\star = \sqrt{\boldsymbol{\alpha}^T\mathbf{\Omega}_\mathbf{r}^{-1}\boldsymbol{\alpha}}$
3. FMPs: Natural instruments for trading in factor space
4. Adding a factor: orthogonalize $\rightarrow$ estimate $\rightarrow$ trade the FMP
5. IR $\approx$ IC $\times \sqrt{nT}$ under the stated assumptions
6. Interpret TC using its stated forecast and risk normalization
7. Signal and portfolio aggregation agree under shared, frictionless linear MVO and matching weights

Learning Objectives

1. When does MVO suggest shorting a positive-return asset?
2. How do constraints address practical considerations?
3. What are the benefits and drawbacks of penalties vs. hard constraints?
4. How do forecast errors affect the population Sharpe ratio of the chosen portfolio?
5. How does the covariance estimation error affect the Sharpe ratio?

Two-Asset Example

Two assets with Sharpe ratios $s_1, s_2 \geq 0$, correlation $-1 < \rho < 1$, and risk scale $\kappa > 0$. Define volatility allocations $v_i = \sigma_i w_i$.

$$\mathbf{C}^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}$$

Optimal volatility allocations:

$$v_1^\star = \frac{\kappa}{1-\rho^2}(s_1 - \rho s_2)$$
$$v_2^\star = \frac{\kappa}{1-\rho^2}(s_2 - \rho s_1)$$

MVO shorts asset 2 when $s_2 < \rho s_1$, even if $s_2 > 0$!
The short hedges exposure to the stronger asset; the allocation can be sensitive to estimation error.

Example: Shorting a Positive-Sharpe Asset: Intuition

From the two-asset solution: $v_2^\star \propto s_2 - \rho\, s_1$. Asset 2 is shorted when $s_2 < \rho\, s_1$.

- **Economic interpretation:** when $s_2 < \rho s_1$, the expected return on asset 2 is too small relative to its exposure to asset 1. A short position in asset 2 can hedge part of that exposure.
- **Is this a feature or a bug?**
  - Feature: mathematically optimal under exact inputs.
  - Bug: estimation error in s_2 or ρ can flip the sign of $v_2^\star$. Since $\partial v_2^\star / \partial s_2 = \kappa/(1-\rho^2)$, errors are amplified as $\rho \to 1$.
- **Thresholds.** With $s_1 = 0.5$ and $\rho = 0.6$, any $s_2 < 0.30$ triggers a short. An estimation error of 0.10 in s_2 can move an estimate near this threshold from one side to the other.

Design principle

If you are uncomfortable with the optimizer shorting positive-Sharpe assets, add a long-only constraint or introduce a lower bound on individual positions. The constraint reduces the feasible set and can improve stability when estimates are noisy; its effect on realized performance depends on the problem.

Impact of Forecast Errors on Sharpe

Suppose estimated Sharpes are $\tilde{s}_i = s_i + \varepsilon_i$ with $|\varepsilon_i| \leq 0.5$.

Figure: $\rho = 0.1$

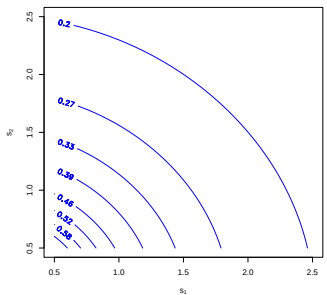
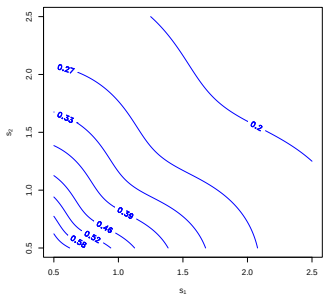


Figure: $\rho = 0.5$



Impact of Correlation Errors

For two assets, let $\mathbf{C}$ be the true correlation matrix with correlation ρ, and let $\widehat{\mathbf{C}}$ use $\hat{\rho} = \rho + \varepsilon$, with $|\hat{\rho}| < 1$. For volatility allocations $\hat{\mathbf{v}}$ chosen using $\widehat{\mathbf{C}}$,

$$\sigma^2_{\text{true}} = \hat{\mathbf{v}}^T \mathbf{C} \hat{\mathbf{v}} = \hat{\mathbf{v}}^T \widehat{\mathbf{C}} \hat{\mathbf{v}} - 2\varepsilon \hat{v}_1 \hat{v}_2.$$

The sign of the difference depends on both the estimation error and the positions.

Figure: $\rho = 0.1$

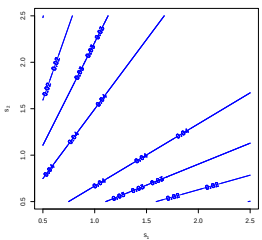
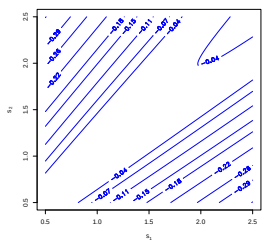
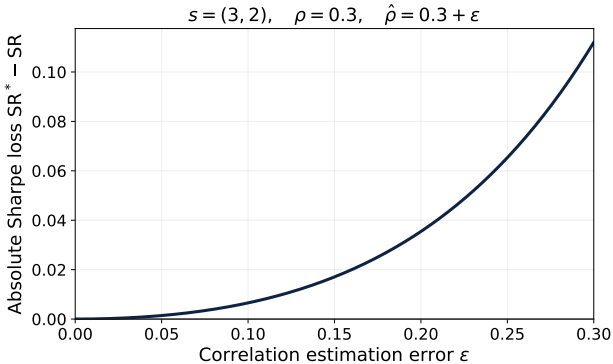


Figure: $\rho = 0.8$



Sharpe Loss vs. Correlation Error



Weights are optimized using $\hat{\rho} = \rho + \varepsilon$ and evaluated using the true correlation ρ. The plot shows the absolute loss from the true optimal Sharpe.

Linear Constraints

Common constraints in portfolio optimization:

- **Long-only:** $w_i \geq 0$.
- **Gross market value:** $\sum_i |w_i| \leq G$. Introduce $\mathbf{w} = \mathbf{x} - \mathbf{y}$, $\mathbf{x}, \mathbf{y} \geq 0$, with $\sum_i (x_i + y_i) \leq G$.
- **Factor exposure:** $\sum_i \beta_i w_i = b_0$ (e.g., market neutrality).
- **Turnover:** $\sum_i |w_i - w_i^{\text{start}}| \leq T_{\max}$; auxiliary variables give a linear formulation.
- **Long/short ratio:** $\sum_i x_i = K \sum_i y_i$ represents the actual ratio only when $x_i y_i = 0$ for every asset. That condition requires known position signs or a nonconvex formulation.

The first four constraints are linear in weights or in an augmented set of variables.

Nonlinear Constraints

- **Trading cost constraint:**

$$\sum_i c_i |w_i - w_i^{\text{start}}|^\gamma \leq C, \quad c_i \geq 0, \ C \geq 0, \ \gamma \in [1, 2]$$

- **Style factor risk constraint:**

$$\mathbf{b}^{\text{style}} = (\mathbf{B}^{\text{style}})^T \mathbf{w}$$
$$(\mathbf{b}^{\text{style}})^T \mathbf{\Omega}_{\mathbf{f}}^{\text{style}} \mathbf{b}^{\text{style}} \leq \sigma_{\text{style}}^2$$

- **Tracking error constraint:**

$$(\mathbf{w} - \mathbf{w}^{\text{bench}})^T \mathbf{\Omega}_{\mathbf{r}} (\mathbf{w} - \mathbf{w}^{\text{bench}}) \leq \sigma_a^2$$

Non-Convex Constraints (Handle with Care)

Maximum number of assets:

$$|w_i| \leq M x_i, \quad x_i \in \{0, 1\}$$
$$\sum_i x_i \leq N_{\max}$$

Here $M > 0$ must be a valid upper bound on every $|w_i|$. This is a mixed-integer program (NP-hard in general).

Problematic constraints:

- Lower bounds on idiosyncratic variance fraction
- Lower bounds on portfolio volatility

These are *non-convex* and can lead to discontinuous, fragile solutions.

Constraints as Penalties (Pricing Out)

Consider a convex optimization problem written as

$$\max_{\mathbf{x}} \; f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) \leq a,$$

where f is concave and g is convex. Under strong duality, an optimal multiplier $\lambda^\star \geq 0$ supports the Lagrangian objective

$$\max_{\mathbf{x}} \; \left[f(\mathbf{x}) - \lambda^\star g(\mathbf{x}) \right].$$

A constrained optimum solves this problem; a penalized solution must also be feasible and satisfy complementary slackness to solve the constrained problem.

Example: diagonal loading

For $\max_{\mathbf{w}} \; \boldsymbol{\alpha}^T\mathbf{w} - \lambda \mathbf{w}^T\boldsymbol{\Omega}\mathbf{w} - \nu\|\mathbf{w}\|^2$, $\lambda > 0$, the risk term uses $\widetilde{\boldsymbol{\Omega}} = \boldsymbol{\Omega} + (\nu/\lambda)\mathbf{I}$. After rescaling, this is shrinkage toward a scaled identity matrix.

Five Approaches to Robust and Stochastic Optimization

Different uncertainty models can lead to regularization penalties. The following forms are illustrative; the result depends on the stated uncertainty set.

Uncertainty	Illustrative penalty	Assumption
Predictive alpha risk	$\lambda\tau^2\|\mathbf{w}\|^2$	added predictive covariance $\tau^2\mathbf{I}$
Robust alpha	$d\|\mathbf{w}\|$	$\|\boldsymbol{\alpha} - \bar{\boldsymbol{\alpha}}\|_2 \leq d$
Missing factor risk	$\lambda\tau^2(\mathbf{b}^T\mathbf{w})^2$	uncorrelated omitted factor with loading $\mathbf{b}$ and variance τ^2
Correlation uncertainty	$\lambda d\|\mathbf{V}\mathbf{w}\|_1^2$	entrywise error bound; conservative penalty
Covariance uncertainty	$\lambda d\|\mathbf{w}\|^2$	Frobenius-norm ball of radius d

Here λ is the risk-penalty coefficient and $\mathbf{V}$ is the diagonal matrix of asset volatilities. Constraints can improve out-of-sample performance, but do not guarantee it.

Robust Alpha

If the true alpha lies in a ball: $\|\boldsymbol{\alpha} - \bar{\boldsymbol{\alpha}}\| \leq d$
For $\mathbf{w} \neq 0$, the worst-case alpha is

$$\boldsymbol{\alpha}^\star = \bar{\boldsymbol{\alpha}} - d\frac{\mathbf{w}}{\|\mathbf{w}\|}.$$

For $\mathbf{w} = 0$, every alpha in the ball gives the same payoff.
Robust MVO $\equiv$ standard MVO with penalty $d\,\|\mathbf{w}\|$:

$$\max_{\mathbf{w}} \bar{\boldsymbol{\alpha}}^T\mathbf{w} - d\,\|\mathbf{w}\| - \lambda\mathbf{w}^T\boldsymbol{\Omega}_\mathbf{r}\mathbf{w}$$

- Penalty is *not* quadratic — it is an ℓ_2 norm
- For $\lambda > 0$ and $\boldsymbol{\Omega}_\mathbf{r} \succ 0$, the optimum is $\mathbf{w} = 0$ if and only if $\|\bar{\boldsymbol{\alpha}}\| \leq d$.

Sharpe Ratio Efficiency (SRE)

Let $\boldsymbol{\alpha}$ and $\boldsymbol{\Omega} \succ 0$ be the true alpha and covariance, and let $\bar{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\Omega}} \succ 0$ be their estimates. For $\hat{\mathbf{w}} = \hat{\boldsymbol{\Omega}}^{-1}\bar{\boldsymbol{\alpha}} \neq 0$ and $\boldsymbol{\alpha} \neq 0$, define

$$\mathrm{SRE} := \frac{\mathrm{SR}_{\mathrm{true}}(\hat{\mathbf{w}})}{\sqrt{\boldsymbol{\alpha}^T\boldsymbol{\Omega}^{-1}\boldsymbol{\alpha}}} = \frac{\boldsymbol{\alpha}^T\hat{\mathbf{w}}}{\sqrt{\hat{\mathbf{w}}^T\boldsymbol{\Omega}\hat{\mathbf{w}}}\,\sqrt{\boldsymbol{\alpha}^T\boldsymbol{\Omega}^{-1}\boldsymbol{\alpha}}}.$$

By Cauchy–Schwarz, $\mathrm{SRE} \leq 1$, with equality if and only if $\boldsymbol{\Omega}^{1/2}\hat{\mathbf{w}}$ is a *positive* multiple of $\boldsymbol{\Omega}^{-1/2}\boldsymbol{\alpha}$. SRE is the geometric transfer coefficient $\mathrm{TC}_{\mathrm{geom}}(\hat{\mathbf{w}})$ defined earlier, with $\boldsymbol{\Omega}$ in place of $\boldsymbol{\Omega}_{\boldsymbol{\epsilon}}$.

Impact of Alpha Error

Assume $\hat{\boldsymbol{\Omega}} = \boldsymbol{\Omega} \succ 0$ and $\boldsymbol{\alpha}, \bar{\boldsymbol{\alpha}} \neq 0$. If

$$\left\| \frac{\boldsymbol{\alpha}}{\|\boldsymbol{\alpha}\|} - \frac{\bar{\boldsymbol{\alpha}}}{\|\bar{\boldsymbol{\alpha}}\|} \right\| \leq \delta_\alpha,$$

then a lower bound on Sharpe ratio efficiency is

$$\mathrm{SRE} \geq 1 - 2 \underbrace{\|\boldsymbol{\Omega}^{-1}\|_{\mathrm{op}} \|\boldsymbol{\Omega}\|_{\mathrm{op}}}_{\text{condition number of } \boldsymbol{\Omega}} \delta_\alpha^2.$$

The bound becomes weaker for an ill-conditioned covariance matrix. It is informative only when its right-hand side is sufficiently close to one.

Impact of Risk Error

Assume the alpha estimate is exact, $\bar{\boldsymbol{\alpha}} = \boldsymbol{\alpha} \neq 0$, and $\boldsymbol{\Omega}, \hat{\boldsymbol{\Omega}} \succ 0$. If

$$\left\| \boldsymbol{\Omega}^{1/2} \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\Omega}^{1/2} - \kappa \mathbf{I} \right\|_{\text{op}} \leq \delta_{\text{risk}}, \qquad 0 \leq \delta_{\text{risk}} < \kappa,$$

then

$$\boxed{\text{SRE} \geq \sqrt{1 - (\delta_{\text{risk}}/\kappa)^2}.}$$

- For $\hat{\boldsymbol{\Omega}} = c\boldsymbol{\Omega}$, $c > 0$, $\kappa = 1/c$ and SRE $= 1$: proportional miscalibration changes scale, not direction.
- If the matrices share eigenvectors, SRE is at least $2\sqrt{\nu_{\min}\nu_{\max}}/(\nu_{\min} + \nu_{\max})$, where $\nu_i = \lambda_i/\hat{\lambda}_i$ for eigenvalues paired in the common basis.

Sharpe Ratio Efficiency: Why the Bounds Hold

Use the operator-2-norm condition number. For alpha error, set $A = \mathbf{\Omega}^{-1/2}$ and let u, v be the unit vectors in the directions of $\boldsymbol{\alpha}, \bar{\boldsymbol{\alpha}}$. For $f(x) = Ax/\|Ax\|$,

$$\|f(u) - f(v)\| \le 2\,\mathrm{cond}(A)\|u - v\|, \qquad \mathrm{cond}(A)^2 = \mathrm{cond}(\mathbf{\Omega}).$$

SRE is $f(u)^T f(v) = 1 - \frac{1}{2}\|f(u) - f(v)\|^2$, which gives the alpha-error bound.
For risk error, let $z = \mathbf{\Omega}^{-1/2}\boldsymbol{\alpha}$ and $M = \mathbf{\Omega}^{1/2}\hat{\mathbf{\Omega}}^{-1}\mathbf{\Omega}^{1/2}$. Then

$$\mathrm{SRE} = \frac{z^T M z}{\|z\|\,\|Mz\|} \ge \frac{2\sqrt{\lambda_{\min}(M)\lambda_{\max}(M)}}{\lambda_{\min}(M) + \lambda_{\max}(M)} \ge \sqrt{1 - (\delta_{\mathrm{risk}}/\kappa)^2}.$$

The middle step is the Kantorovich inequality; the last uses $\lambda(M) \in [\kappa - \delta_{\mathrm{risk}}, \kappa + \delta_{\mathrm{risk}}]$.

Robust Portfolio Construction: Key Takeaways

1. MVO can short positive-return assets to hedge correlated exposures.
2. Forecast and covariance errors can change optimal weights and reduce their population Sharpe ratio.
3. Constraints and penalties can moderate sensitivity to estimation error, but their effects depend on the model and the true parameters.
4. Sharpe ratio efficiency measures the loss in portfolio direction relative to the true optimum.
5. Proportional risk miscalibration changes portfolio scale but leaves its Sharpe ratio unchanged.

Solutions to Quizzes

Quiz-I Heuristic Error

For positive-definite $\mathbf{C}$ and $\mathbf{s} \neq 0$, let $\mathbf{v}_{\mathrm{h}} = \mathbf{s}$ and $\mathbf{v}^{\star} = \mathbf{C}^{-1}\mathbf{s}$. Their Sharpe ratio comparison is

$$\frac{\mathrm{SR}(\mathbf{v}_{\mathrm{h}})}{\mathrm{SR}^{\star}} = \frac{\mathbf{s}^{T}\mathbf{s}}{\sqrt{(\mathbf{s}^{T}\mathbf{C}\mathbf{s})(\mathbf{s}^{T}\mathbf{C}^{-1}\mathbf{s})}} \leq 1.$$

The loss is one minus this ratio. For the allocation vectors,

$$\frac{\|\mathbf{v}^{\star} - \mathbf{v}_{\mathrm{h}}\|}{\|\mathbf{v}^{\star}\|} = \frac{\|(\mathbf{I} - \mathbf{C})\mathbf{v}^{\star}\|}{\|\mathbf{v}^{\star}\|} \leq \|\mathbf{I} - \mathbf{C}\|_{\mathrm{op}}\,.$$

The last bound concerns unnormalized allocations, not Sharpe loss.

Quiz-I Heuristic Error: Eigenvalue Bound

For a symmetric correlation matrix $\mathbf{C}$ with eigenvalues λ_i,

$$\|\mathbf{I} - \mathbf{C}\|_{\mathrm{op}} = \max_i |1 - \lambda_i|.$$

For positive-definite $\mathbf{C}$ of dimension $n \geq 2$, $\sum_i \lambda_i = n$ and $0 < \lambda_i < n$, so $\|\mathbf{I} - \mathbf{C}\|_{\mathrm{op}} < n - 1$.

- If $\mathbf{C} = \mathbf{I}$, the heuristic and optimizer agree in direction.
- A large eigenvalue or a small eigenvalue can increase the discrepancy.

Quiz-II: Correlations, Dispersion, and Limits to Performance I

For $n \geq 2$, let $\mathbf{C} = (1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T$. Its eigenvalues are $1 + (n-1)\rho$ along $\mathbf{1}$ and $1-\rho$ on its orthogonal complement. Hence

$$-\frac{1}{n-1} \leq \rho \leq 1.$$

The inverse below requires strict inequalities; the endpoints are singular.

Quiz-II: Correlations, Dispersion, and Limits to Performance II

For $\bar{s} = n^{-1}\sum_i s_i$ and $\mathrm{var}(\mathbf{s}) = n^{-1}\sum_i (s_i - \bar{s})^2$,

$$\mathbf{C}^{-1} = \frac{1}{1-\rho}\mathbf{I} - \frac{\rho}{(1-\rho)[1+(n-1)\rho]}\mathbf{1}\mathbf{1}^T,$$

$$\mathbf{v}^\star = \mathbf{C}^{-1}\mathbf{s} = \frac{1}{1-\rho}\left(\mathbf{s} - \frac{n\rho}{1+(n-1)\rho}\bar{s}\,\mathbf{1}\right).$$

$$\mathrm{SR}^{\star 2} = \mathbf{s}^T\mathbf{C}^{-1}\mathbf{s} = \frac{n}{1-\rho}\,\mathrm{var}(\mathbf{s}) + \frac{n\bar{s}^2}{1+(n-1)\rho}.$$

If $s_i = s$ for every asset, $\mathrm{SR}^\star = |s|\sqrt{n/[1+(n-1)\rho]}$. For fixed $0 < \rho < 1$, this tends to $|s|/\sqrt{\rho}$ as $n \to \infty$; for $\rho = 0$ it grows as $|s|\sqrt{n}$.

Quiz-III: The Correlation Chain Rule

Suppose X, Y, Z have finite, nonzero variances, X and Z are conditionally independent given Y, and both $E[X \mid Y]$ and $E[Z \mid Y]$ are affine functions of Y. Then

$$\mathrm{Corr}(X, Z) = \mathrm{Corr}(X, Y)\,\mathrm{Corr}(Y, Z).$$

Indeed, conditional independence gives $\mathrm{Cov}(X, Z) = \mathrm{Cov}(E[X \mid Y], E[Z \mid Y])$; affine conditional means give the product formula after standardization. Conditional independence alone is insufficient: if $Y \sim N(0, 1)$ and $X = Z = Y^2$, the two correlations with Y are zero but $\mathrm{Corr}(X, Z) = 1$.

Quiz-IV: Hit Rate and IC

Let α be a scalar forecast and r the associated return. Assume (α, r) are jointly Gaussian, both centered, with positive variances. After dividing by their standard deviations, write

$$(r, \alpha) \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right), \qquad \rho = \mathrm{Corr}(\alpha, r).$$

Positive rescaling does not change signs or correlation.

Hit Rate

A hit occurs when the forecast and return have the same sign:

$$h = P(r \geq 0, \alpha \geq 0) + P(r < 0, \alpha < 0) = P(r\alpha > 0).$$

For $|\rho| < 1$, central symmetry makes these two quadrant probabilities identical. If $x = (r, \alpha)^T$ and $\mathbf{\Sigma}$ is the correlation matrix,

$$h = \frac{2}{2\pi\sqrt{\det \mathbf{\Sigma}}} \int_0^\infty \int_0^\infty \exp\Big(-\tfrac{1}{2} x^T \mathbf{\Sigma}^{-1} x\Big)\, dr\, d\alpha. \tag{1}$$

The singular cases $\rho = \pm 1$ follow by continuity.

Multivariate Normal distribution of α and returns

We plot the multivariate distribution of α (x-axis) and security returns (y-axis). The hit rate is the probability of the shaded first and third quadrants. Equation 1 integrates over the first quadrant and doubles it, by central symmetry.

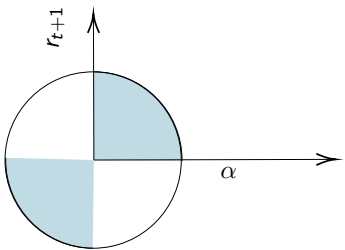


Figure: Case1: Correlation $\rho = 0$

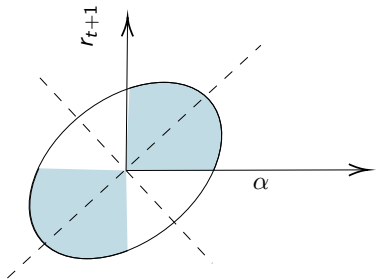


Figure: Case2: Correlation $\rho > 0$

From the symmetry of Figure 9 it is obvious that the hit rate is 0.5 or 50%. We calculate a closed form solution for Case 2.

Hit Rate: Whitening and Polar Coordinates

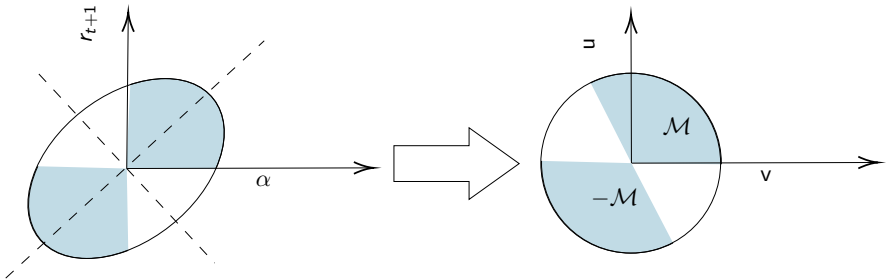
Let U, V be independent standard normal variables and set

$$r = U, \qquad \alpha = \rho U + \sqrt{1 - \rho^2}\, V.$$

In polar coordinates, $U = R \sin\theta$, $V = R \cos\theta$. The radius is Rayleigh distributed and the angle is uniform on $[0, 2\pi)$, independently of R. The same-sign region consists of two opposite wedges, each with angle $\frac{\pi}{2} + \arcsin\rho$.

Transformations

Figure: Whitening transforms the ellipse into a circle



Let $\mathcal{M}$ denote the upper wedge. By symmetry,

$$h = \frac{1}{\pi}\left(\iint_{\mathcal{M}} \exp\left(-\frac{u^2+v^2}{2}\right) dudv\right) \tag{1}$$

Polar transformation of the integral

$$h = \frac{1}{\pi}\left(\int_{R=0}^{\infty} \exp\left(-\frac{R^2}{2}\right) RdR\right)\left(\int_{\theta=0}^{\pi/2+\arcsin\rho} d\theta\right) \tag{2}$$

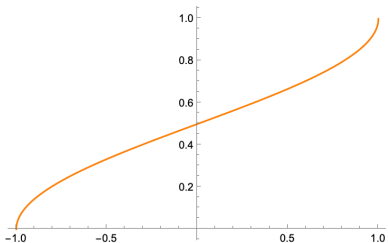
The boundary of the quadrant is obtained by setting:

$$\begin{aligned} \alpha &= R(\rho\sin\theta + \sqrt{1-\rho^2}\cos\theta) = 0 \\ \theta &= \frac{\pi}{2} + \arcsin\rho \end{aligned} \tag{3}$$

The first integral is one. Therefore,

$$h = \frac{1}{2} + \frac{\arcsin\rho}{\pi} \tag{4}$$

Figure: Hit rate vs. correlation



Expected Maximum Drawdown for Brownian Motion with Drift

For $X_t = \mu t + \sigma W_t$ with $\mu > 0$, let

$$\bar{D}_T := \sup_{0 \le u \le v \le T} (X_u - X_v), \qquad \text{SR} := \mu/\sigma, \qquad \alpha := \text{SR}\sqrt{T/2}.$$

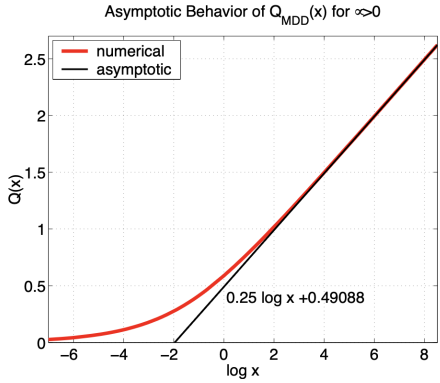
Then (Magdon-Ismail et al.2004)

$$\begin{aligned} E\left[\bar{D}_T\right] &= \frac{2\sigma^2}{\mu} Q_{\bar{D}}(\alpha^2) \\ &= \frac{2\sigma}{\text{SR}} Q_{\bar{D}}\left(\frac{\text{SR}^2\ T}{2}\right) \\ Q_{\bar{D}}(x) &= \tfrac{1}{4}\ln x + c + o(1), \quad c \approx 0.49088, \quad x \to \infty \end{aligned}$$

Expected Maximum Drawdown: Example

For annualized $\sigma = 10\%$, SR $= 2.5$ and $T = 1$ year, $x = 3.125$. Interpolating the paper's table gives $Q_{\bar{D}}(x) \approx 0.82$ and $E[\bar{D}_T] \approx 6.6\%$. This is an arithmetic drawdown of X, not a percentage drawdown of an exponential wealth process.

Figure: Asymptotic behavior of $Q_{\bar{D}}(x)$



Key Portfolio Management Terms I

Gross Market Value (GMV) For portfolio value V and weights w_i, the sum of absolute position values: $\text{GMV} = V \sum_i |w_i|$.

Net Market Value (NMV) Sum of signed position values: $\text{NMV} = V \sum_i w_i$.

Dollar Volatility Standard deviation of portfolio P&L in dollar terms, rather than as a percentage return.

Idiosyncratic Risk The contribution of residual returns to portfolio variance under a specified factor model.

Factor Risk The contribution of common factors to portfolio variance under a specified factor model. Factor and residual variances add when factor and residual returns are uncorrelated; their volatilities do not generally add.

Sharpe Ratio Risk-adjusted return: $\text{SR} = \dfrac{E[R_p] - R_f}{\sigma(R_p - R_f)}$.

Close vs. Arrival A comparison of the closing price with the arrival price; execution quality can separately be measured against either benchmark.

Leverage / Gross Leverage GMV/NAV; exposure run per dollar of capital.

Key Portfolio Management Terms II

Marginal Contribution to Risk (MCR) The derivative of portfolio volatility with respect to a position: $\mathrm{MCR}_i = (\mathbf{\Omega_r w})_i / \sigma_p$ when $\sigma_p > 0$.

Risk/Variance Decomposition Splitting total portfolio variance into factor and idiosyncratic components.

Information Ratio Mean active return (portfolio return minus benchmark return) divided by tracking error, the standard deviation of active return.

Transfer Coefficient A measure of alignment between forecasts and positions; its exact definition and risk normalization depend on the model.

Information Coefficient (IC) Correlation between predicted alpha and realized returns; a signal-quality measure.

VWAP Slippage Execution price vs. volume-weighted average price over the trading horizon.

Implementation Shortfall Cost relative to the decision-price paper portfolio, including delay, execution, and the opportunity cost of unfilled orders. Arrival-price benchmarks exclude pre-arrival delay.

Market Impact Price movement caused by one's own trading activity.

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